LUMAFLOW · PROJECT PLAN

LumaFlow project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lumaflow		
Source	docs/11-plan.md @ f88a9c3		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Reactor tracer test added to P4 verification (O-4 (a), decided by Amish 2026-10-25)
0.1	2026-10-25	Amish Chadha	Kit 1.8 upgrade: workstreams, resources, stage gates and references

LumaFlow project plan

LumaFlow is at TRL 3 and pipeline stage P3 (design outputs), the portfolio cap. The next gate, P4 verification, is on hold until Amish lifts the cap; the decision requested now is the concept selection between options A and B (LMF-DEC-001, O-1). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and requirements	Project lead (Amish)	PRB, URS, PRS, OPT, concept selection	Done on paper; O-1 awaiting Amish
Science and calculations	Design lead	STM, CAL, RTM	Done; STEM check open on four measurement items
Design outputs	Design lead	Model at D3 for every option, STEP, drawings, priced BOM, build plan, DEC	Done for option A
Regulatory and risk	Design lead with a consultant later	REG, RSK with design FMEA	First pass done; many rows "to assess"
Partner	Project lead (Amish)	Agreement with a university extension programme for private well owners	Not yet approached
Verification (P4)	Design lead with the partner	Test plans and lab reports, including the reactor tracer test for the residence-time distribution (O-4, decided by Amish 2026-10-25)	On hold (TRL cap 3)

Resources needed

Table 2. Resources (for P4, on hold; costs are rough estimates for planning, not quotes).

Resource	Deliverable	Effort	Estimated cost (USD)
Prototype parts	One unit to the build plan	Parts as BOM	433 (priced BOM)
Machine shop	Two end caps, tube boss weld, holder	1 to 2 days of shop time	Included in the BOM estimates
UV-C reflectance and transmittance measurement	Liner reflectance (S8) and transmittance (S10) at 275 nm	Partner or university optics lab	To assess
Biodosimetry (MS2 or similar)	Measured RED at 90 %/cm and 70 %/cm	Partner microbiology lab	To assess
Firmware sketch and bench rig	Dose rule, 1 s fail-safe, interlocks (PR-004)	Electronics lead, about 2 weeks	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Met (kit 1.8 upgrade, 2026-10-25)
P1 Definition	URS, PRS draft, RSK, MKT	Met
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR	Met on paper; the DDR is proposed, awaiting Amish
P3 Design outputs	Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability	Met for option A (2026-10-25)
P4 Verification	Test plans and lab reports, verification report	On hold (TRL cap 3)
P5 Validation	Field trial with the partner on real wells	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. **Amish:** choose between option A (axial UV-C LED reactor) and option B (hollow-fibre ultrafilter), which scored within 10 % (O-1), and whether to graft the runner-up's best idea (O-2).
2. **Amish:** send the first message to the extension programme (draft in the review note of 2026-10-02).
3. When the TRL cap is lifted: measure the liner (S8, S10), run biodosimetry (PR-002, PR-003), write the firmware sketch and test PR-004, then plate counts for regrowth and photorepair (S6, S7).

References

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